Vibration Damping of an Aluminum Ring

Lab Day 2

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Abstract

This experiment investigates the application of passive piezoelectric shunt damping to a clamped aluminum ring. An impact hammer and Frequency Response Function (FRF) analysis were used to characterize the system's modal properties. Piezoelectric patches on the ring were grouped into three 135 nF batches (A, B, and C). By comparing open-circuit and short-circuit FRFs, the electromechanical coupling factor (K_c) was calculated for five modes for each batch. This analysis identified promising candidates for damping, notably Mode 1 for Batch A ($K_c \approx 0.053$) and Batch C ($K_c \approx 0.075$), and Mode 3 for Batch C ($K_c \approx 0.042$). Based on these results, two passive shunt circuits were implemented: one targeting Mode 3 of Batch C, and a second targeting Mode 1 by combining Batches A and C in parallel. Results confirmed the theoretical principle that doubling the patch capacitance halves the required tuning inductance for a target frequency. This study successfully demonstrates the complete process of modal identification, coupling factor calculation, and practical application of tuned piezoelectric damping.

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1 Introduction

Vibrations in mechanical structures are a common source of noise, material fatigue, and potential system failure. Passive damping techniques are often sought for their reliability and simplicity, as they do not require external power or complex control systems. One such method is piezoelectric (PZT) shunt damping, which utilizes the piezoelectric effect to convert mechanical strain energy into electrical energy, where it can be dissipated by a passive electrical circuit.

This method relies on bonding PZT patches to a structure. These patches function as capacitors. When the structure vibrates at a natural frequency, the patches are strained, generating a voltage. By connecting an external "shunt" circuit typically an inductor (L) and resistor (R) to the PZT patch, an **RLC** resonant circuit is formed. When this electrical circuit is tuned to a target structural vibration frequency, it can effectively dissipate large amounts of energy, thereby damping the structural mode.

The effectiveness of this damping is critically dependent on the electromechanical coupling factor (K_c), which represents the system's ability to convert mechanical energy to electrical energy at a specific mode. A higher K_c factor indicates a stronger coupling and a greater potential for damping.

The objective of this experiment is to apply this principle to a clamped aluminum ring. First, the ring's modal properties are characterized using FRF analysis to identify modes with a high coupling factor. Second, based on this analysis, optimal passive shunt circuits are designed and implemented. This report details the process of identifying target modes from three PZT batches (A, B, and C) and presents the results of two damping attempts: a single-batch circuit targeting Mode 3 (Batch C) and a combined-batch circuit targeting Mode 1 (Batches A and C).

2 Methodology

2.1 Theoretical Principles and Formulas

The methodology is based on calculating the system's modal parameters to design an optimal passive shunt circuit. The key parameters are the electromechanical coupling factor (K_c), the optimal tuning inductance (L_{opt}), and the optimal tuning resistance (R_{opt}).

These are derived by comparing the modal properties of the structure in two-boundary electrical conditions: open-circuit (oc) and short-circuit (sc), specifically their resonant frequencies (ω_{open} and ω_{short}).

2.1.1 Electromechanical Coupling Factor (K_c)

This factor is a critical measure of the PZT patch's ability to convert mechanical energy into electrical energy at a given mode. A high K_c is essential for effective damping. It is calculated from the frequency shift:

$$K_c^2 = \frac{\omega_{\text{open}}^2 - \omega_{\text{short}}^2}{\omega_{\text{short}}^2}$$

2.1.2 Optimal Shunt Parameters

Once a target mode with a high K_c is identified, the shunt circuit components are calculated. The **optimal tuning inductance (L_{opt})** is tuned to the open-circuit frequency of the target mode ($\omega_0 = \omega_{\text{open}}$) and the clamped capacitance of the PZT batch (C_{eps}):

$$L_{\text{opt}} = \frac{1}{C_{\text{eps}}\,\omega_{\text{open}}^2}$$

The **optimal tuning resistance (R_{opt})** for maximizing energy dissipation is then calculated based on the coupling factor and capacitance:

$$R_{\text{opt}} = \frac{\sqrt{3/2}\cdot K_c}{C_{\text{eps}}\,\omega_{\text{open}}}$$

2.2 Experimental Procedure and Data Acquisition

2.2.1 Apparatus

A clamped aluminum ring with 15 piezoelectric patches was used, as shown in Figure 1. The 15 patches were grouped into 3 batches (A, B, and C). Each batch consisted of 4 piezoelectric patches, with each of the 3 batches totaling 135 nF.

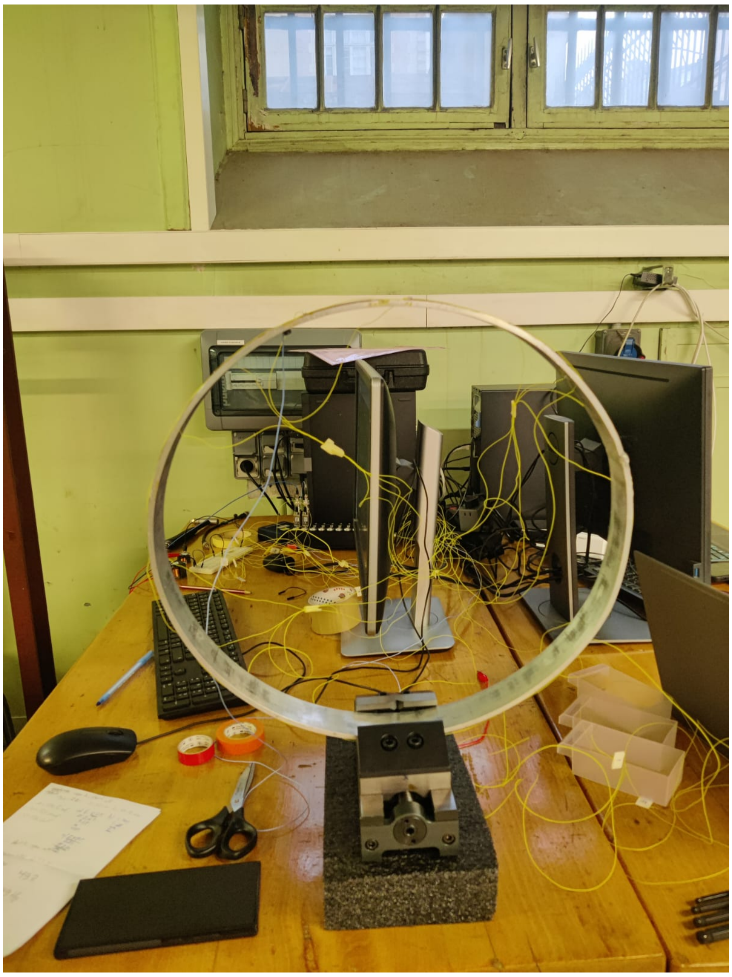

Figure 1: The clamped aluminum ring experimental setup, showing the ring, piezoelectric patches, and impact hammer.

2.2.2 FRF Measurement

An impact hammer, along with DewesoftX software, was used to acquire the Frequency Response Function (FRF) for all short-circuit and open-circuit conditions for batches A, B, and C separately.

2.2.3 MATLAB Analysis

The raw FRF data from DewesoftX was imported into MATLAB for analysis (see Appendix B). The MATLAB code was used to:

- Load the open-circuit and short-circuit '.mat' files.
- Perform modal fitting ('modalfit') to extract the precise resonant frequencies (f_{open}, f_{short}) and damping ratios (ζ_{open}, ζ_{short}) for the first five modes.
- Apply the formulas from Section 3.1 to calculate the K_c factor, L_{opt}, and R_{opt} for each mode and batch.
- The results of this analysis are summarized in Table 1.

2.3 Damping Strategy

The results from the MATLAB analysis (presented in Table 1) guided the selection of damping targets. The primary goal was to find modes with a high K_c factor, as these offer the most damping potential.

The analysis identified three promising candidates:

- **Mode 1, Batch C ($K_c \approx 0.075$)**
- **Mode 1, Batch A ($K_c \approx 0.053$)**
- **Mode 3, Batch C ($K_c \approx 0.042$)**

Based on these results and the availability of lab components, two damping strategies were executed:

1. **Damping C3:** A single-batch shunt targeting Mode 3 of Batch C. Although C1 had a higher K_c, C3 was chosen due to the practical availability of the required $\approx 6.5\,\text{H}$ inductor.
2. **Damping AC1:** A combined-batch shunt targeting Mode 1. Since both A and C had good coupling at this mode, they were connected in **parallel**. This strategy was also intended to validate the theory that doubling the capacitance ($C_{\text{eps,AC}} \approx 270\,\text{nF}$) would halve the required tuning inductance.

The results of implementing these two strategies are presented in the following section.

3 Data and Analysis

3.1 Initial FRF Analysis

The data obtained from the MATLAB analysis, which formed the basis for the damping strategy, is presented in Table 1.

Table 1: FRF Analysis Results (Rounded to 2 Significant Figures)

Pair Label	**Mode**	f_{open} (Hz)	ζ_{open}	f_{short} (Hz)	ζ_{short}	k_c **factor**	L (H)	R (Ω)
Open_A_vs_Short	1	50	0.013	50	0.016	0.053	75.0	1500.0
Open_A_vs_Short	2	120	0.08	120	0.087	0.027	14.0	330.0
Open_A_vs_Short	3	170	0.08	170	0.087	0.031	6.5	260.0
Open_A_vs_Short	4	260	0.08	260	0.087	—	—	—
Open_A_vs_Short	5	470	0.08	470	0.087	0.01	0.84	31.0
Open_B_vs_Short	1	50	0.013	50	0.016	0.019	99.0	740.0
Open_B_vs_Short	2	120	0.077	120	0.087	0.016	18.0	260.0
Open_B_vs_Short	3	170	0.077	170	0.087	—	—	—
Open_B_vs_Short	4	260	0.077	260	0.087	—	—	—
Open_B_vs_Short	5	470	0.077	470	0.087	—	—	—
Open_C_vs_Short	1	50	0.014	50	0.016	0.075	76.0	2200.0
Open_C_vs_Short	2	120	0.038	120	0.087	0.028	14.0	350.0
Open_C_vs_Short	3	170	0.038	170	0.087	0.042	6.48	360.0
Open_C_vs_Short	4	260	0.038	260	0.087	—	—	—
Open_C_vs_Short	5	470	0.038	470	0.087	—	—	—

3.2 Damping C3 Results

The first strategy targeted Mode 3 of Batch C. Figure 2 shows the comparison between the undamped FRF (open- and short-circuit conditions) and the three different damping configurations tested.

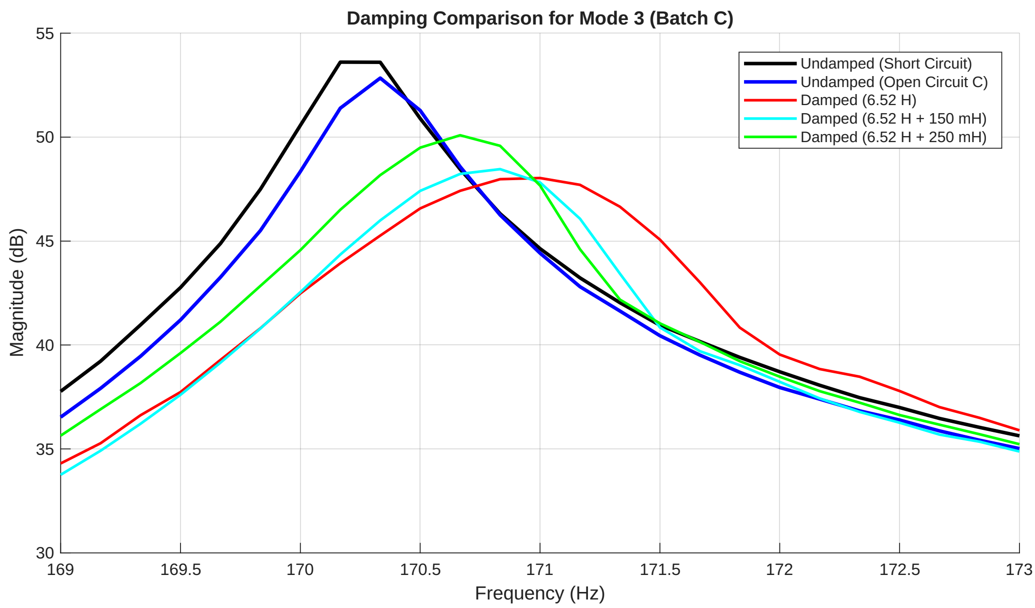

Figure 2: Damping comparison for Batch C, Mode 3.

The plot shows that the configuration with the single 6.52 H inductor produces the *strongest* damping effect: its resonance peak is the lowest among all the curves. In addition, this peak is shifted slightly to the right (towards higher frequency) compared to the undamped case. This indicates that the electrical loading not only adds damping but also slightly modifies the effective dynamic stiffness seen by the mode.

When the additional 150 mH and 250 mH inductors are placed in series with the 6.52 H coil, the total inductance increases, but the damping effect is actually reduced: the corresponding peaks are higher than for 6.52 H alone. This suggests that these larger equivalent inductances move the electrical resonance away from the optimal tuning given approximately by

$$L_{\mathrm{opt}} = \frac{1}{C_{\mathrm{eps}}\,\omega_0^2},$$

so that less energy is transferred from the mechanical mode into the shunt circuit.

Overall, the results confirm two important points: (i) a correctly tuned shunt (here, the 6.52 H case) can significantly reduce the resonance amplitude, and (ii) even small deviations from the optimal inductance value reduce the damping and can slightly shift the apparent resonant frequency.

Three configurations were tested using a 6.4 H (labeled) inductor, which measured 6.52 H at 170Hz:

- The 6.52 H inductor alone.
- The 6.52 H inductor in series with a 150 mH (labeled) inductor.
- The 6.52 H inductor in series with a 250 mH (labeled) inductor.

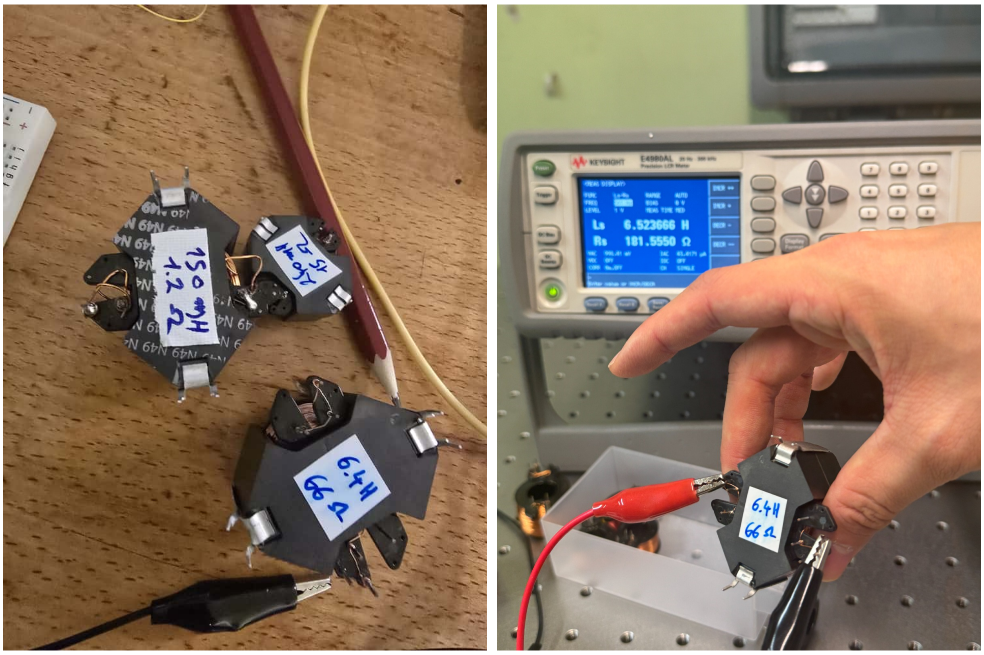

Figure 3: The inductors used for the C3 damping experiment. **Left:** The set of inductors with their labeled values (6.4 H, 150 mH, and 250 mH). **Right:** The 6.4 H inductor (labeled) being measured on an LCR meter, confirming the measured value of 6.52 H as mentioned in the text.

3.3 Damping AC Results

As both patches A and C showed good K_c factors for Mode 1, we connected them in **parallel** as planned. In theory, this doubles the total capacitance ($C_{\mathrm{eps,AC}} \approx 2 \times 135\,\mathrm{nF}$), which halves the required tuning inductance. We connected a 34 H inductor to the combined batch AC.

Figure 4 shows the FRF comparison between the undamped short-circuit condition and the shunt applied to the combined A and C batches connected in parallel.

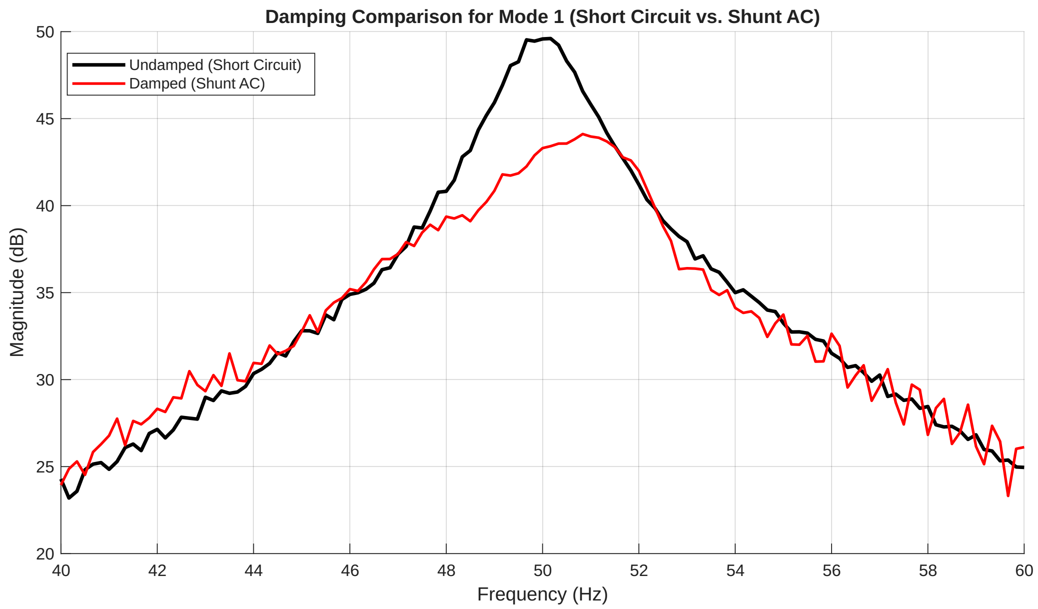

Figure 4: Damping comparison for Batch AC on Mode 1.

The plot indicates that the shunt connected to the combined AC batch successfully reduces the amplitude of the Mode 1 resonance. The peak magnitude decreases noticeably relative to the short-circuit case, confirming that connecting the A and C patches in parallel increases the effective capacitance.

As predicted by theory, doubling the capacitance through parallel connection requires approximately half the inductance compared to using a single batch alone. The measured response supports this: with a 34 H inductor, the electrical anti-resonance aligns closely with the structural resonance, leading to visible damping.

4 Conclusion

This experiment successfully demonstrated the application of **passive piezoelectric shunt damping** to a clamped aluminum ring, following a rigorous process of modal characterization and circuit design.

The initial Frequency Response Function (FRF) analysis, comparing open-circuit and short-circuit conditions across the three piezoelectric batches (A, B, and C), allowed for the critical determination of the **electromechanical coupling factor (K_c)**. The highest coupling was found for:

- **Mode 1** of Batch C ($K_c \approx 0.075$)
- **Mode 1** of Batch A ($K_c \approx 0.053$)
- **Mode 3** of Batch C ($K_c \approx 0.042$)

Based on component availability, two distinct damping implementations were executed.

1. Single-Batch Damping (C3): Targeting Mode 3 of Batch C ($f_0 \approx 170\,\text{Hz}$) with an optimally tuned inductance of $L_{\text{opt}} \approx 6.48\,\text{H}$. The measured 6.52 H inductor produced the **strongest reduction** in the resonance peak, confirming the theoretical principle of a tuned shunt circuit. Deviations from this optimal value significantly reduced the damping effectiveness, highlighting the **sensitivity** of the method to accurate tuning.

2. Combined-Batch Damping (AC1): Targeting Mode 1 ($f_0 \approx 50\,\text{Hz}$) by connecting Batches A and C in **parallel**. This parallel configuration effectively doubled the total clamped capacitance ($C_{\text{eps,AC}} = 2 \times 135\,\text{nF} = 270\,\text{nF}$). The successful application of a 34 H inductor, which is approximately half the calculated $\approx 75\,\text{H}$ for a single batch, **validated the theoretical scaling principle** that L_{opt} is inversely proportional to C_{eps}, and successfully reduced the amplitude of the Mode 1 resonance.

In summary, the experiment confirmed the importance of a high K_c factor and the critical need for precise tuning of the shunt inductance for maximum vibration suppression.

The required optimal tuning inductance (L_{opt}) for the targeted low-frequency modes (50–170 Hz) was extremely high, ranging from 6.48 H to 76 H (for a single batch at Mode 1). The formula for optimal inductance is $L_{\text{opt}} = (C_{\text{eps}}\,\omega_0^2)^{-1}$. Passively achieving such large inductances for low-frequency applications is highly challenging, typically requiring custom-wound coils with thousands of turns around a high-permeability ferromagnetic core, or a **synthetic inductor circuit** (Gyrator) to simulate the required impedance actively. This high-inductance requirement is the primary practical hurdle in implementing passive PZT shunt damping.

4.1 Practical Inductor Design for Low-Frequency Damping

As noted, the primary hurdle for this damping technique is achieving the extremely high optimal inductance values (L_{opt}), which ranged from 6.48 H to 76 H for the low-frequency modes. The provided course material on inductor design explains how such components are constructed and what design challenges they present, particularly for achieving high inductance for piezoelectric damping.

A passive inductor consists of a **coil of copper wire** wrapped around a **coil former**, which is then enclosed in a **core of magnetic material** (like ferrite). The final inductance L is determined by the formula:

$$L = A_L N^2$$

where N is the **number of turns** in the coil and A_L is the **permeance** of the magnetic core.

To achieve a very high inductance (e.g., 76 H), this formula shows we must either have a core with a very high A_L value or a very large number of turns N. Manufacturers like TDK provide a wide range of cores, such as the **RM series**, with different materials offering different permeance values. For example, an "ungapped" RM 10 core using T38 material provides an A_L of 16 000 nH. To achieve $L = 76\,\text{H}$ with this core, the number of turns required would be:

$$N = \sqrt{\frac{L}{A_L}} = \sqrt{\frac{76\,\text{H}}{16\,000 \times 10^{-9}\,\text{H}}} \approx 2180 \text{ turns}$$

This large number of turns introduces a significant practical trade-off: **parasitic resistance**. Real inductors have a **series resistance (R_s)** which models the copper losses from the wire. This resistance is calculated as:

$$R_s = \rho \frac{N l_N}{S_w}$$

where ρ is the resistivity of copper, l_N is the mean length of one turn, and S_w is the wire's cross-sectional area.

To quantify this problem, we can estimate the R_s for the 76 H inductor. Using the filling factor formula $k_u = \frac{NS_w}{A_N}$ and assuming standard specifications for an RM 10 core ($A_N \approx 44.2\,\text{mm}^2$, mean turn length $l_N \approx 52\,\text{mm}$) and a k_u of 0.5, the required wire cross-section S_w would be:

$$S_w = \frac{k_u A_N}{N} = \frac{0.5 \times (44.2 \times 10^{-6}\,\text{m}^2)}{2180} \approx 1.01 \times 10^{-8}\,\text{m}^2$$

This is an extremely thin wire (diameter $\approx 0.11\,\text{mm}$). Using the standard resistivity for copper ($\rho \approx 1.72 \times 10^{-8}\,\Omega \cdot \text{m}$), the parasitic resistance R_s is:

$$R_s = \rho \frac{N l_N}{S_w} = (1.72 \times 10^{-8}\,\Omega \cdot \text{m}) \frac{2180 \times (0.052\,\text{m})}{1.01 \times 10^{-8}\,\text{m}^2} \approx 193\ \Omega$$

This calculated parasitic resistance of $\approx 193\,\Omega$ is not negligible. For the C1 mode, the target optimal resistance is $2200\,\Omega$ (see Table 1). While this R_s is only about 9% of the optimal value in this high-inductance case, for other modes (like C3, where $R_{opt} = 360\,\Omega$), a poorly optimized inductor could have a parasitic resistance that is a significant fraction of the total required tuning resistance.

Therefore, designing a custom inductor for this application is an optimization problem. One must select a core with a high A_L value to minimize the required turns N, and then use the largest possible wire diameter that the filling factor and winding area will allow, in order to keep the parasitic series resistance R_s to a minimum. This R_s is critical, as it contributes to the total resistance R in the shunt circuit and affects the damping performance.

5 Out of the box: Advanced Solutions with Synthetic Inductors

Our experiment and the analysis in the previous section highlighted the primary practical limitation of passive shunt damping: the requirement for **extremely large and impractical passive inductors** [2]. The provided literature offers two advanced solutions to this problem by replacing the physical coil with an active electronic circuit called a **synthetic inductor**, or **gyrator** [3]. This circuit typically uses operational amplifiers (OpAmps), resistors, and capacitors to mimic the impedance of an ideal inductor, thus solving the problem of size, mass, and high parasitic resistance associated with large wirewound coils.

However, simply replacing the passive coil with a standard gyrator, such as the common **Antoniou circuit**, introduces two new, significant real-world challenges: (1) tunability and (2) voltage limitations [1].

5.1 Challenge 1: Adaptive Tuning with VCSI

The shunt circuits we implemented were *passive*, meaning they are tuned to a single, fixed frequency. In real-world applications, a structure's resonance frequencies can shift due to environmental factors (like temperature) or changes in loading. A fixed shunt would then become "mistuned" and ineffective.

The solution is an **adaptive shunt** that can adjust its inductance to track the structure's changing resonance frequency [3]. This is achieved using a **Voltage-Controlled Synthetic**

Inductor (VCSI). This is a gyrator circuit where one of the key resistors, which sets the inductance value, is replaced with a **voltage-controlled resistor (VCR)**.

The literature [3] presents two modern approaches for creating a linear, high-performance VCR:

- **Using an Analog Multiplier:** An analog multiplier, such as the **AD633**, can be configured to act as a VCR. This approach was found to have enhanced linearity over a limited voltage control range.
- **Using an OTA:** An **Operational Transconductance Amplifier (OTA)**, such as the **LM13700**, can also be used as a VCR.

Both methods create a synthetic inductor whose inductance value can be changed dynamically by an external control voltage, enabling a "smart" damping system that remains optimal even when the structure's properties change [3].

5.2 Challenge 2: High-Voltage Operation

A second, critical limitation of standard OpAmp-based gyrators is their low voltage range [1]. While our lab experiment used low-amplitude impacts, real-world mechanical structures especially as they become lighter and more flexible can experience high vibration levels. These vibrations can induce voltages across the piezoelectric (PEM) electrodes that are **well over** 100 V [1].

Standard OpAmps saturate around 30 V **peak-to-peak (Vpp)**, and even specialized high-voltage OpAmps like the **OPA445** or the **OPA454** are limited to a 90 V to 100 V peak-to-peak range. This is insufficient for many applications.

To solve this, a **novel high-voltage synthetic inductor** design is proposed [1]. This circuit uses the 100 V peak-to-peak OPA454 OpAmp as its base but combines two techniques to dramatically extend its range:

1. **Output Voltage Boost Configuration:** The OpAmp's own power supply rails are dynamically adjusted by two auxiliary "boost" OpAmps, effectively doubling the output range to 200 V peak-to-peak.
2. **Bridge Amplifier Configuration:** Two of these 200 V peak-to-peak "boosted" gyrator circuits are then driven anti-phase, with the load (the piezo) connected between them.

By combining these two methods, the circuit which uses a total of six OPA454 OpAmps can effectively **quadruple the range to** 400 V **peak-to-peak** [1]. This robust design allows synthetic inductors to be used for effective vibration damping even in high-amplitude, real-world industrial or aerospace applications where standard circuits would fail [1].

A FRF Plots

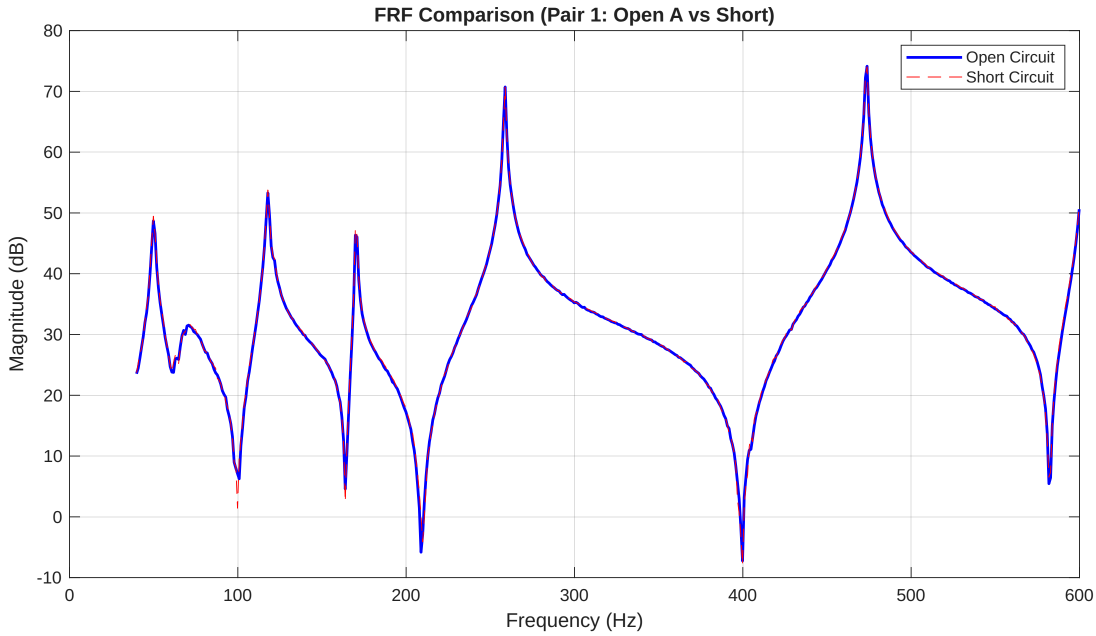

Figure 5: Open-circuit vs short-circuit FRF for Batch A.

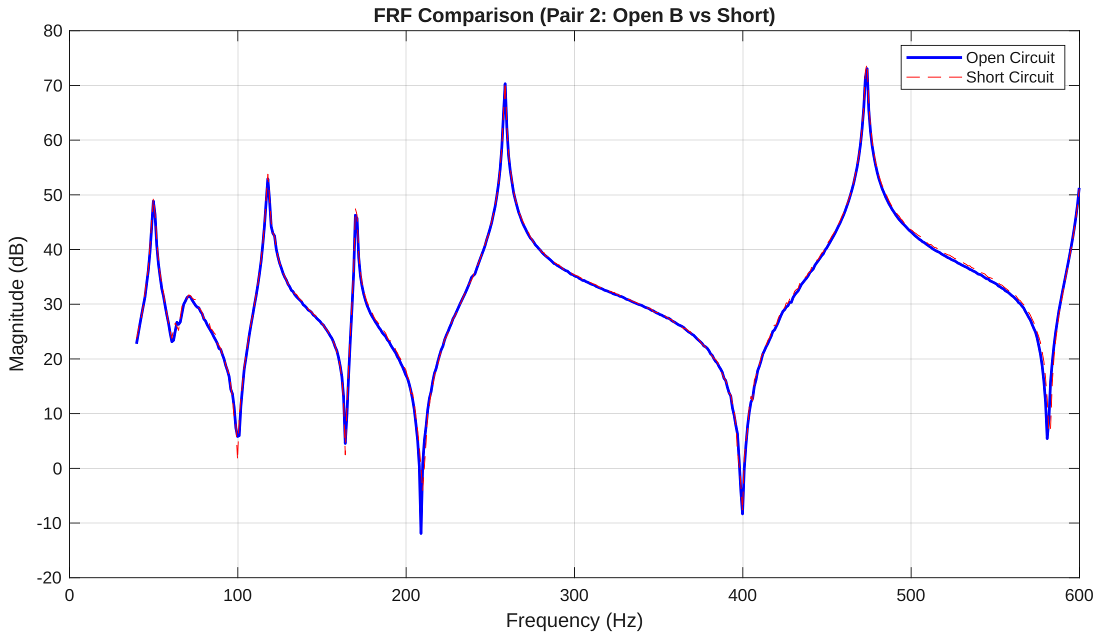

Figure 6: Open-circuit vs short-circuit FRF for Batch B.

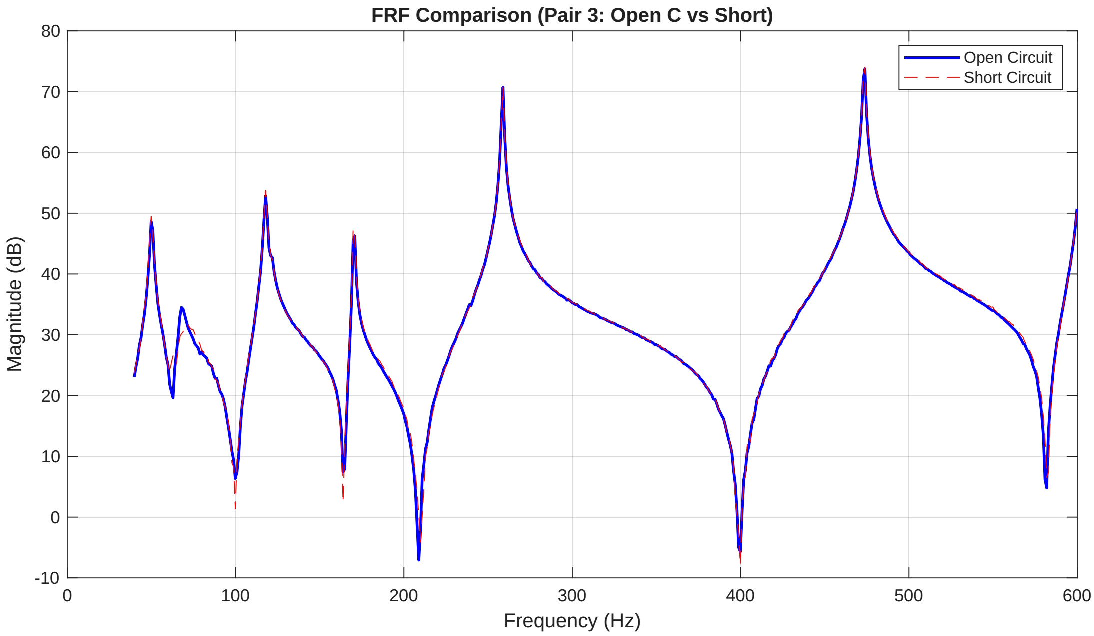

Figure 7: Open-circuit vs short-circuit FRF for Batch C.

B MATLAB Code

The following MATLAB code was used to analyze the FRF and obtain the data.

```
% Clear environment
clc
clear all
close all

% --- 1. Define General Parameters and Data Sets ---
idx_range = (40:600)*6;
fs = 200; % Sample frequency (Used in modalfit)
num_modes = 5; % Number of modes to identify

% --- File Pairs ---
file_pairs = {
    {'Test_mxj_open_new_a.mat', 'Test_mxj_short_new.mat'},
    {'Test_mxj_open_new_b.mat', 'Test_mxj_short_new.mat'},
    {'Test_mxj_open_new_c.mat', 'Test_mxj_short_new.mat'}
};

modes_to_analyze = 1:num_modes; % This will be 1:5
max_model_order = 12; % High order for robust LSRF damping calculation

% --- 2. Initialize Results Storage ---
% We will store all results in a cell array first
all_results = cell(0, 11); % 0 rows, 11 columns
column_names = { ...
    'Pair_Label', 'Mode', 'Ceps_nF', 'f_open_Hz', 'zeta_open', ...
    'f_short_Hz', 'zeta_short', 'kc_factor', 'L_H', 'R_Ohm', 'Warning' ...
};


% --- 3. Start Main Loop for all File Pairs ---
for pair_idx = 1:length(file_pairs)
    open_file = file_pairs{pair_idx}{1};
    short_file = file_pairs{pair_idx}{2};

    % --- Ceps Assignment Logic ---
    pair_label_str = ''; % For the results table
    if contains(open_file, '_a.mat')
        Ceps = 135e-9; % 135 nF for set 'a'
        pair_label_str = 'Open_A_vs_Short';
    elseif contains(open_file, '_b.mat')
        Ceps = 135e-9; % 103 nF for set 'b'
        pair_label_str = 'Open_B_vs_Short';
    elseif contains(open_file, '_c.mat')
        Ceps = 135e-9; % 133 nF for set 'c'
        pair_label_str = 'Open_C_vs_Short';
    else
        fprintf('Warning: No Ceps rule matched for %s. Using 135nF fallback.\n', open_file);
        Ceps = 135e-9;
        pair_label_str = ['Unknown_Pair_' num2str(pair_idx)];
    end

    fprintf('\n=======================================================\n');
    fprintf('            ANALYSIS FOR DATA PAIR %d: %s\n', pair_idx, pair_label_str);
    fprintf('  Open Circuit Data: %s\n  Short Circuit Data: %s\n', open_file, short_file);
```

```
    fprintf('  Ceps (Clamped Capacitance) = %.2f nF\n', Ceps * 1e9);
    fprintf('=======================================================\n');

    % --- 4. Process Open Circuit Data ---
    try
        FRF_O = load(open_file);
        idx_range_O = idx_range(idx_range <= length(FRF_O.Data1_X_MT_FRF_H1_2Zplus_1Zplus_Real));
        if isempty(idx_range_O)
            fprintf('Warning: idx_range too large for %s. Skipping pair.\n', open_file);
            continue;
        end
        freq_O = FRF_O.Data1_X_MT_FRF_H1_2Zplus_1Zplus_Real(idx_range_O);
        real_part_O = FRF_O.Data1_MT_FRF_H1_2Zplus_1Zplus_Real(idx_range_O);
        imag_part_O = FRF_O.Data1_MT_FRF_H1_2Zplus_1Zplus_Imag(idx_range_O);
        func_O = real_part_O + 1i * imag_part_O;

        % --- 5. Process Short Circuit Data ---
        FRF_S = load(short_file);
        idx_range_S = idx_range(idx_range <= length(FRF_S.Data1_X_MT_FRF_H1_2Zplus_1Zplus_Real));
        if isempty(idx_range_S)
            fprintf('Warning: idx_range too large for %s. Skipping pair.\n', short_file);
            continue;
        end
        freq_S = FRF_S.Data1_X_MT_FRF_H1_2Zplus_1Zplus_Real(idx_range_S);
        real_part_S = FRF_S.Data1_MT_FRF_H1_2Zplus_1Zplus_Real(idx_range_S);
        imag_part_S = FRF_S.Data1_MT_FRF_H1_2Zplus_1Zplus_Imag(idx_range_S);
        func_S = real_part_S + 1i * imag_part_S;
    catch e
        fprintf('Error loading MAT file: %s. Skipping pair.\n', e.message);
        continue;
    end

    % --- 6. Plot Both FRFs ---
    figure(pair_idx);
    plot(freq_O, 20*log10(abs(func_O)), 'b', 'LineWidth', 1.5);
    hold on;
    plot(freq_S, 20*log10(abs(func_S)), 'r--');
    hold off;
    title(sprintf('FRF Comparison (Pair %d: %s)', pair_idx, pair_label_str));
    xlabel('Frequency (Hz)');
    ylabel('Magnitude (dB)');
    legend('Open Circuit', 'Short Circuit');
    grid on;

    % --- 7. Perform Modal Fit (HYBRID STRATEGY) ---
    fprintf('Performing Step 1: Peak-Picking to anchor frequencies...\n');
    f_anchor_O = modalfit(imag(func_O), freq_O, fs, num_modes, 'FitMethod', 'PP');
    f_anchor_S = modalfit(imag(func_S), freq_S, fs, num_modes, 'FitMethod', 'PP');

    f_anchor_O = f_anchor_O(:);
    f_anchor_S = f_anchor_S(:);

    fprintf('Performing Step 2: LSRF fit using anchored frequencies...\n');
    [~, zeta_O] = modalfit(imag(func_O), freq_O, fs, num_modes, ...
                            'FitMethod', 'LSRF', 'Order', max_model_order, ...
```

```
                            'PhysFreq', f_anchor_O);
    [~, zeta_S] = modalfit(imag(func_S), freq_S, fs, num_modes, ...
                            'FitMethod', 'LSRF', 'Order', max_model_order, ...
                            'PhysFreq', f_anchor_S);
    f_0_O = f_anchor_O;
    f_0_S = f_anchor_S;

    disp('--- Identified Modal Parameters ---');
    fprintf('Open Circuit (f_O): '); disp(f_0_O');
    fprintf('Short Circuit (f_S):'); disp(f_0_S');
    fprintf('Open Circuit (zeta_O): '); disp(zeta_O');
    fprintf('Short Circuit (zeta_S):'); disp(zeta_S');

    % --- 8. Calculate and Store Parameters for each Mode (1 to 5) ---
    for mode_index = modes_to_analyze
        fprintf('\n--- RESULTS FOR MODE %d ---\n', mode_index);

        % Initialize all result variables to NaN or empty
        f_mode_O = NaN; om_mode_O = NaN; zeta_mode_O = NaN;
        f_mode_S = NaN; om_mode_S = NaN; zeta_mode_S = NaN;
        kc_factor = NaN; L_mode = NaN; R_mode = NaN;
        warning_msg = '';

        % Robust check if the mode was successfully identified
        if mode_index > length(f_0_O) || mode_index > length(f_0_S)
            fprintf('Warning: Mode %d was not identified. Skipping.\n', mode_index);
            warning_msg = 'Mode not identified by Peak-Picking';
        else
            % Get frequencies and damping for the current mode
            f_mode_O = f_0_O(mode_index);
            om_mode_O = 2 * pi * f_mode_O;
            zeta_mode_O = zeta_O(mode_index);

            f_mode_S = f_0_S(mode_index);
            om_mode_S = 2 * pi * f_mode_S;
            zeta_mode_S = zeta_S(mode_index);

            fprintf(' Resonant Frequencies & Damping:\n');
            fprintf('  Open (f_O): %.2f Hz | (om_O): %.2f rad/s | (zeta_O): %.4f\n', f_mode_O, om_mode_O, zeta_mode_O);
            fprintf('  Short (f_S): %.2f Hz | (om_S): %.2f rad/s | (zeta_S): %.4f\n', f_mode_S, om_mode_S, zeta_mode_S);

            om_o = om_mode_O;
            om_s = om_mode_S;

            if om_o > om_s
                % kc = sqrt( (om_open^2 - om_short^2) / om_short^2 )
                kc_factor = sqrt( (om_o^2 - om_s^2) / om_s^2 );
                % L = 1 / (Ceps * om_open^2)
                L_mode = 1 / (Ceps * om_o^2);
                % R = sqrt(3/2) * kc / (Ceps * om_open)
                R_mode = sqrt(3/2) * kc_factor / (Ceps * om_o) ;

                fprintf(' Electromechanical Coupling Factor (kc%d): %.4f\n', mode_index, kc_factor);
                fprintf(' Equivalent Circuit Parameters:\n');
                fprintf('  Inductance (L%d): %.2f H\n', mode_index, L_mode);
                fprintf('  Resistance (R%d): %.2f Ohms\n', mode_index, R_mode);
            else
```

```matlab
                fprintf(' Warning: Short-circuit omega (%.2f rad/s) is not less than open-circuit omega (%.2f rad/s).\n', om_s, om_o);
                fprintf(' kc%d and R cannot be reliably calculated for this mode.\n', mode_index);
                warning_msg = 'om_s >= om_o';
                % L can still be calculated
                L_mode = 1 / (Ceps * om_o^2);
                fprintf('  Inductance (L%d): %.2f H (Calculated independently)\n', mode_index, L_mode);
            end
        end

        % --- Add all data to the results cell array ---
        result_row = { ...
            pair_label_str, ...
            mode_index, ...
            Ceps * 1e9, ...
            f_mode_O, ...
            zeta_mode_O, ...
            f_mode_S, ...
            zeta_mode_S, ...
            kc_factor, ...
            L_mode, ...
            R_mode, ...
            warning_msg ...
        };
        all_results = [all_results; result_row]; % Append the new row

    end % End of mode_index loop
end % --- End Main Loop ---

% --- 9. Convert to Table and Save to Excel ---
disp('\n*** Full Analysis Complete ***\n');

if ~isempty(all_results)
    % Convert cell array to a table
    results_table = cell2table(all_results, 'VariableNames', column_names);

    % Display table in command window
    disp('--- Summary of All Results ---');
    disp(results_table);

    % Define filename and save
    excel_filename = 'day_2_analysis.xlsx';
    try
        writetable(results_table, excel_filename, 'Sheet', 'Modal_Analysis_Results', 'WriteMode', 'replacefile');
        fprintf('\nSuccessfully saved results to:\n%s\n', fullfile(pwd, excel_filename));
    catch e
        fprintf('\n--- ERROR SAVING EXCEL FILE ---\n');
        fprintf('Error: %s\n', e.message);
        fprintf('If you are on MATLAB Online, check file permissions.\n');
        fprintf('If you are on desktop, make sure the file is not open.\n');
    end
else
    disp('No results were generated. Excel file not created.');
end
```

Listing 1: MATLAB code for FRF analysis

References

[1] K. Dekemele, P. Van Torre, and M. Loccufier. High voltage synthetic inductor for vibration damping in resonant piezoelectric shunt. *Journal Title*, XX(X):1–10, 2020.

[2] B. Lossouarn and P. Flament. Smart structures: Piezoelectric shunt damping, session 5. In *International Master: Structural Mechanics and Coupled Systems*, 2025. Presentation.

[3] M. Vatavu, V. Nastasescu, F. Turcu, and I. Burda. Voltage-controlled synthetic inductors for resonant piezoelectric shunt damping: A comparative analysis. *Applied Sciences*, 9(22): 4777, 2019.